

TOPIC 5 MAGNETISM & ELECTROMAGNETIC INDUCTION

Answer all the questions

1.

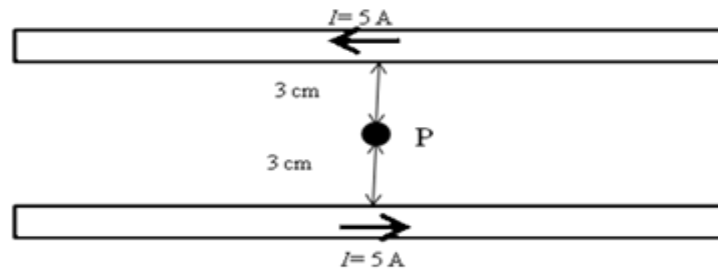


FIGURE 5.1

FIGURE 5.1 shows two long parallel wires, each carrying a 5 A current in opposite direction. Determine the magnetic field, $\mathbf{B}$ at point P, 3 cm from both wires.

[ANS: $6.66 \times 10^{-5}\text{ T}$]

2. The length of a 500 turns solenoid is 3.0 cm. Calculate the magnetic field intensity inside the solenoid when it carries a 0.6 A current.

[ANS: 0.013 T]

3. Two long parallel wires, P and Q are fixed on a horizontal plane north-south direction. Wire P is 10.0 cm left of Q. Wire P carries 40 A current northwards while wire Q carries 20 A current southwards. Determine the magnitude and direction of the resultant magnetic field at midpoint between the wires.

[ANS: $2.4 \times 10^{-4}\text{ T}$: upward]

4.

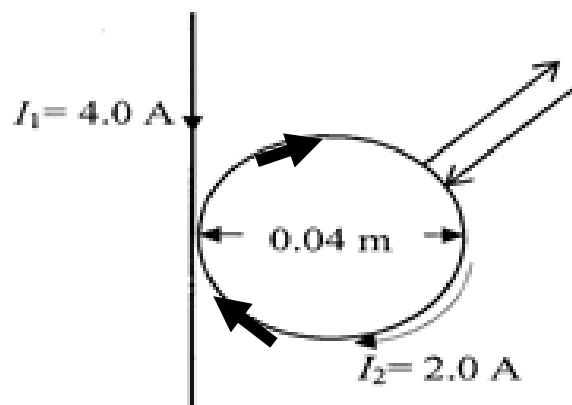


FIGURE 5.2

FIGURE 5.2 shows a long straight wire carrying a current of $I_1 = 4.0\text{ A}$ is placed to the left of a circular loop. The loop has a diameter of $d = 0.04\text{ m}$ and carrying a current of $I_2 = 2.0\text{ A}$. Determine the magnitude and direction of the net magnetic field at the center of the loop.

[ANS: $-2.28 \times 10^{-5}\text{ T}$: into the page]

5.

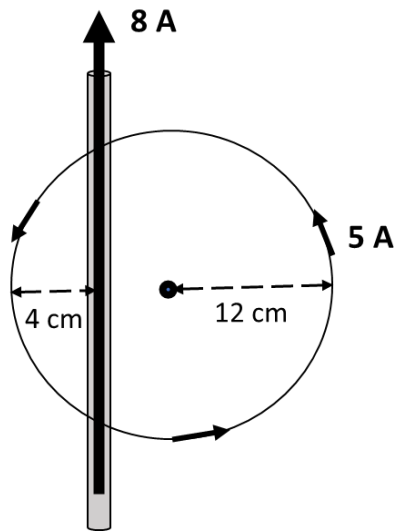


FIGURE 5.3

FIGURE 5.3 shows a vertical straight wire inside a horizontal single loop. The radius of the loop is 12.0 cm and the wire is 4.0 cm from the loop. The currents in the wire and the loops are 8.0 A and 5.0 A, respectively. Determine the magnitude and direction of the resultant magnetic field at the centre of the loop

[ANS: $0.62 \times 10^{-5} \text{ T}$: out of the page]

6. A 2.0 m long wire is made into a plane coil of five turns. If the currents in the wire is 30 mA. Calculate the magnetic field at the centre of the coil.

[ANS: $2.96 \times 10^{-7} \text{ T}$]

7.

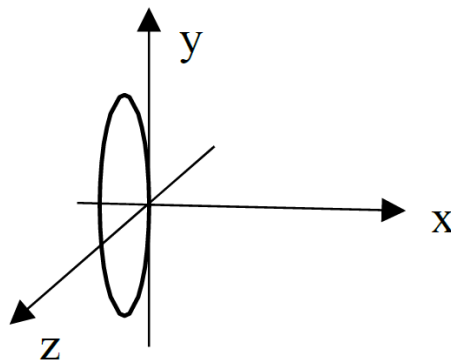


FIGURE 5.4

FIGURE 5.4 shows the plane of a circular coil of radius 2.0 m is placed in a uniform magnetic field. The magnetic field is 0.40 T and in the positive x-direction. Calculate the magnetic flux through the circular coil.

[ANS: 5.03 Wb]

TOPIC 5 | REINFORCEMENT QUESTIONS DP024

8. A small search coil with 50 turns and cross-sectional area $4.5 \times 10^{-4} \text{ m}^2$ is placed in a uniform magnetic field of flux density 2.8 T. The plane of the coil is perpendicular to the field.
- Calculate the magnetic flux linked with the coil.
 - The magnetic field is reduced to zero in 0.2 s. Calculate the e.m.f induced in the coil.

[ANS: 0.063 Wb, 0.315 V]

9. A metal rod moves perpendicularly through a 1.5 T uniform magnetic field at a speed of 2.0 cm s^{-1} . If the length of the rod is 40.0 cm and the resistance is 3Ω . Calculate
- the induced emf.
 - the induced current.

[ANS: 0.012 V, $4 \times 10^{-3} \text{ A}$]

10.

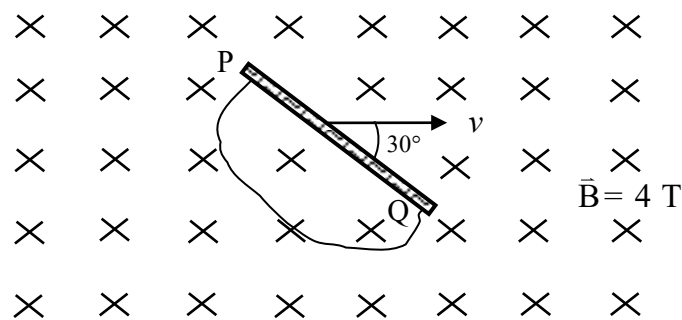


FIGURE 5.5

A metal rod PQ of length 0.2 m is pulled to the right in a uniform magnetic field of 4 T with velocity 0.5 m s^{-1} as shown in **FIGURE 5.5**. If the resistance of the rod is 10Ω and the velocity make an angle 30° with the length of the rod, calculate

- the induced emf in the rod.
- the induced current.

[ANS: 0.4 V, 0.04 A]